

SER216 Final Practical Project

Digital Parking Permit System — Final Report

Student: Shahzod Nematov · **Course:** SER216 — Software Engineering · **Term:** Spring 2026 · **Date:** 2026-05-04

Repository: github.com/Shahzod1602/ser216_Final_Project_ShahzodNematov

Scenario

A university wants a digital parking permit system. Students and staff can register vehicles, request parking permits, check available parking zones, and report parking issues. Security staff can verify permits and update violation records.

Task 1 — Requirements

Functional Requirements

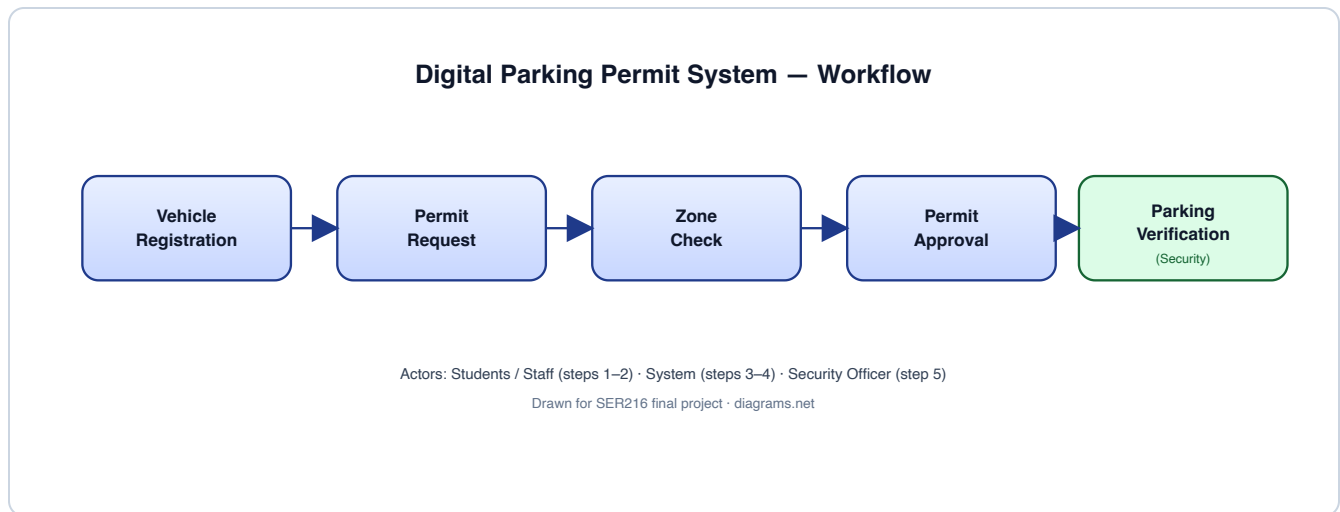
1. **FR-1 Vehicle Registration.** A student or staff member shall be able to register a vehicle by entering plate number, make, model, color, and owner ID. Duplicate plate numbers are rejected.
2. **FR-2 Permit Request.** A registered user shall be able to submit a parking permit request, choosing zone (A/B/C/Visitor) and period (semester, annual, daily). The system returns a confirmation number.
3. **FR-3 Zone Availability Check.** The system shall display real-time availability for each parking zone (free spots, capacity, and status: open / full / closed) before the user picks a zone.
4. **FR-4 Permit Verification by Security.** Security staff shall be able to scan or enter a permit ID and immediately see whether the permit is valid, expired, suspended, or unknown, with the linked vehicle and owner.

Non-Functional Requirements

1. **NFR-1 Performance.** Permit verification at security gates shall return a result within **2 seconds** for 95% of requests under a load of 200 concurrent verifications.
2. **NFR-2 Security & Privacy.** All personal data (owner ID, plate, address) shall be transmitted over TLS 1.2+, stored encrypted at rest, and access shall be limited by role (student, staff, security, admin).

Task 2 — Workflow Diagram

End-to-end workflow drawn in diagrams.net (source: `diagrams/workflow_diagram.svg`):



Task 3 — Test Case Table

Stored in `tables/test_cases.csv`. Six cases — two system, two acceptance, two regression.

Test ID	Feature	Input / Data	Expected Result	Test Type
TC-01	Permit Request Flow	Registered user requests a Zone A semester permit; chosen zone has free capacity	Permit request saved, status Pending Approval , confirmation number returned	System
TC-02	Permit Verification	Security scans valid permit ID linked to a registered vehicle at an open gate	System shows VALID , owner name, plate, and expiry within 2 seconds	System
TC-03	Student Self-Service	Student logs in and registers a new vehicle with valid plate, make, model, color	Vehicle appears in <i>My Vehicles</i> ; user confirms the flow is intuitive (UAT pass)	Acceptance
TC-04	Security UAT	Security officer verifies 10 different permits during a shift drill	All results displayed clearly within 2 seconds; officer confirms workflow works	Acceptance
TC-05	Duplicate Plate Guard	After login fix, register a vehicle with a plate that already exists	System rejects the registration with the same error message as before the fix	Regression
TC-06	Permit Expiry Display	After defect fix, verify a permit whose expiry date is in the future	Permit shows VALID (no longer wrongly EXPIRED)	Regression

System testing checks the integrated end-to-end behaviour. Acceptance testing checks that real users (students and security staff) agree the system meets their expectations and the business rules.

Task 4 — Defect Analysis

Defect: "A valid parking permit is shown as expired during security verification."

Item	Analysis
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Defect category	Functional defect — incorrect business-logic output (date / time-zone calculation bug in permit verification).
Severity	High. A valid permit holder is wrongly flagged as expired, blocking legitimate users from parking and forcing manual security overrides. Not Critical (no crash) but breaks a core feature.
Priority	High / P1. Must be fixed in the next release. The defect runs on the security-gate path many times per day and is highly visible.
Possible root cause	Expiry comparison most likely uses the server's local date while <code>expiry_date</code> is stored in UTC (or vice versa). Around the day boundary, today's date is still $\leq$ expiry in UTC but already $>$ expiry after a naive local-date cast. Other candidates: off-by-one on inclusive/exclusive end date, or stale cached permit status not refreshed after renewal.
Regression test	TC-06: after the fix, verify a permit whose <code>expiry_date</code> is in the future across multiple time-zone boundaries (just before midnight local, just after midnight local, and on the exact expiry day). Expected: status VALID in all three runs. Add this test to the regression suite so the bug cannot return.

Task 5 — GitHub Evidence

All seven evidence artefacts are present in the repository:

#	Requirement	Where
1	Repository created	github.com/Shahzod1602/ser216_Final_Project_ShahzodNematov
2	README.md	Repo root
3	Feature branch	<code>feature/test-plan</code>
4	≥ 3 commits	Commit history on the feature branch and on <code>main</code> after merge
5	Pull request	PR #1, <code>feature/test-plan</code> → <code>main</code>
6	Pull request merged	PR #1 — purple "Merged" badge
7	Final PDF uploaded	<code>final_report.pdf</code> in repo root

Repository homepage



main Branches Tags

Go to file

Code

5 Commits

- diagrams
- notes
- tables
- README.md
- final_report.md

README

SER216 Final Project — Digital Parking Permit System

Student: Shahzod Nematov Course: SER216 — Software Engineering Term: Spring 2026 Submission Date: 2026-05-04

Project Overview

A digital parking permit system for a university. The system lets students and staff register vehicles, request parking permits, check available parking zones, and report parking issues. Security staff verify permits and update violation records.

The repository holds every artifact required for the final practical exam: requirements, workflow diagram, test plan, defect analysis, and the supporting GitHub evidence (branches, commits, pull request, merge).

Tools Used

Purpose	Tool
Report writing	Google Docs
Test case table	Google Sheets
Workflow diagram	diagrams.net
Version control	Git + GitHub
PDF generation	macOS textutil / Preview

Files in this Repository

```
ser216_Final_Project_ShahzodNematov/  
├── README.md           - this file  
├── final_report.pdf     - final report submitted with the project  
├── final_report.md      - markdown source of the report  
├── diagrams/  
│   └── workflow_diagram.svg - Vehicle Registration → ... → Verification  
├── tables/  
│   └── test_cases.csv     - 6 test cases (system, acceptance, regression)  
├── screenshots/  
│   ├── repo_homepage.png  
│   ├── readme.png  
│   ├── commits.png  
│   ├── branch.png  
│   ├── pull_request.png  
│   ├── pr_merged.png  
│   └── pdf_uploaded.png  
├── notes/  
└── tools_used.md
```

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Report repository

Releases

No releases published

Packages

No packages published

Contributors 1



Shahzod1602 Shahzod Nematov

README.md rendered



Files

main

Go to file

- diagrams
- notes
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```

Workflow

The work followed the workflow taught in SER216:

1. Capture the requirements (4 functional + 2 non-functional).
2. Draw the end-to-end workflow diagram in diagrams.net.
3. Build the test case table in Google Sheets (6 cases — 2 system, 2 acceptance, 2 regression).
4. Analyze the supplied defect (expired permit shown for a valid permit).
5. Produce the final PDF report and push everything to GitHub through a feature branch and pull request.

Commit history (≥ 3 commits)



Commits

main

All users

All time

Commits on May 4, 2026

Merge pull request #1 from Shahzod1602/feature/test-plan

Shahzod1602 authored 5 minutes ago

Verified

fcf1688



Add test case table

Shahzod1602 committed 8 minutes ago

aa7f021



Add workflow diagram

Shahzod1602 committed 8 minutes ago

5ede1a2



Add project report structure and tooling notes

Shahzod1602 committed 8 minutes ago

52cb134



Initial commit

Shahzod1602 authored 21 minutes ago

Verified

f26843b



Feature branch



Branches

Overview Active Stale All

Search branches...

Default

Branch	Updated	Check status	Behind	Ahead	Pull request
main	5 minutes ago				...

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
feature/test-plan	8 minutes ago				...

Pull request



Digital Parking Permit System — final project artefacts #1

Merged Shahzod1602 merged 3 commits into main from feature/test-plan 5 minutes ago

Conversation Commits 3 Checks Files changed

+331 -1



Shahzod1602 commented 7 minutes ago

Owner

Summary

- Add the full SER216 final-practical write-up: requirements, workflow, test plan, defect analysis.
- Add the workflow diagram (SVG) and the 6-case test plan (CSV).
- Update the README so it describes the project, tools, files, and workflow.

Tasks covered

- ☒ Task 1 — 4 functional + 2 non-functional requirements
- ☒ Task 2 — workflow diagram (diagrams/workflow_diagram.svg)
- ☒ Task 3 — 6 test cases, 2 each of system / acceptance / regression
- ☒ Task 4 — defect analysis for the "valid permit shown as expired" bug
- ☒ Task 5 — GitHub evidence (this branch, these commits, this PR, the merge)

Test plan

- ☒ Test case table reviewed in tables/test_cases.csv
- ☒ Diagram opens cleanly in the browser
- ☒ README renders correctly on GitHub

Reviewers

No reviews

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

1 participant



Shahzod1602 added 3 commits 8 minutes ago

Add project report structure and tooling notes 52cb134

Add workflow diagram 5ede1a2

Add test case table aa7f021

Shahzod1602 merged commit fcf1668 into main 5 minutes ago

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Pull request merged



Digital Parking Permit System — final project artefacts #1

Merged Shahzod1602 merged 3 commits into `main` from `feature/test-plan` 5 minutes ago

Conversation 0

Commits 3

Checks 0

Files changed 5

+331 -1



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Add project report structure and tooling notes

52cb134

Add workflow diagram

5ede1a2

Add test case table

aa7f021

Shahzod1602 merged commit `fcf1668` into `main` 5 minutes ago

Reviewers

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Assignees

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Labels

None yet

Projects

None yet

Milestone

No milestone

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None yet

1 participant



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Final PDF uploaded in repository

[screenshot pending: PDF uploaded in repo]

Submission Checklist

- ✓ 4 functional + 2 non-functional requirements
- ✓ Workflow diagram (Vehicle Registration → Permit Request → Zone Check → Permit Approval → Parking Verification)
- ✓ 6 test cases (2 system, 2 acceptance, 2 regression)
- ✓ Defect analysis (category, severity, priority, root cause, regression test)
- ✓ Repository created on GitHub
- ✓ README.md describing project, tools, files, workflow
- ✓ Feature branch `feature/test-plan`
- ✓ ≥ 3 meaningful commits
- ✓ Pull request opened and merged
- ✓ Final PDF uploaded with repository link inside the report

— End of report —